Automated Medical Abstract Classification Using Deep Learning

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1. Introduction

The volume of biomedical literature has grown at a pace that far exceeds the capacity of clinicians and researchers to review it manually. PubMed alone indexes more than 1.5 million new articles each year, creating significant challenges for information organization and retrieval in both clinical and research environments. Therefore, efficiently routing newly published research to the appropriate clinical specialties or research groups is an increasingly important task for healthcare institutions, pharmaceutical companies, and medical libraries. Automated classification systems capable of assigning disease categories to medical abstracts could substantially reduce the time required for literature triage, improve knowledge discovery, and enable more scalable biomedical information management.

From a technical perspective, medical abstract classification presents a challenging natural language processing (NLP) problem. Biomedical writing contains dense domain-specific terminology, specialized abbreviations, and subtle semantic distinctions between related diseases. Traditional bag-of-words models can capture highly distinctive medical vocabulary but lack contextual understanding, while modern transformer architectures such as BERT can model deeper semantic relationships through contextual embeddings. A critical research question in biomedical NLP is thus whether domain-specific pretraining on biomedical corpora provides measurable advantages over general-purpose language models when applied to medical text.

In this project, we develop and compare several approaches for automated medical abstract classification across five disease categories: *Neoplasms*, *Digestive System Diseases*, *Nervous System Diseases*, *Cardiovascular Diseases*, and *General Pathological Conditions*. Our models range from classical machine learning baselines to modern transformer architectures, allowing us to isolate the impact of increasingly sophisticated representations. In particular, we investigate whether BiomedBERT, a transformer model pretrained exclusively on biomedical literature, provides a meaningful performance advantage over general-domain models such as BERT.

1.1 Dataset

We evaluate our models using the TimSchopf/medical_abstracts dataset, publicly available on Hugging Face [4]. The dataset contains 14,438 labeled medical abstracts drawn from peer-reviewed biomedical literature. It comprises 11,550 training samples and 2,888 test samples, with each abstract assigned one of five disease categories corresponding to major

medical domains. Abstract lengths range from approximately 170 to 4,000 characters, reflecting the natural variability of clinical research writing [1].

A key characteristic of the dataset is class imbalance, which presents an additional modeling challenge. The largest class, *General Pathological Conditions*, contains 3,844 training samples, more than three times the size of the smallest class, *Digestive System Diseases* (1,195 samples). Without appropriate mitigation strategies, classifiers may disproportionately favor the majority class, resulting in degraded performance on minority categories. Table 1 summarizes the class distribution across the training and test splits.

Class	Training Samples	Testing Samples
Neoplasms	2,530	633
Digestive System Diseases	1,195	299
Nervous System Diseases	1,540	385
Cardiovascular Diseases	2,441	610
General Pathological Conditions	3,844	961
Total	11,550	2,888

Table 1: Dataset label distribution across training and testing splits

2. Methods

We implement four models in a deliberate progression from classical machine learning to domain-specific deep learning, thereby isolating the contribution of each design choice.

2.1 TF-IDF + Logistic Regression (Baseline)

As a classical baseline, we represent each abstract using Term Frequency–Inverse Document Frequency (TF-IDF) features and train a logistic regression classifier. TF-IDF is a widely used representation for text classification that converts documents into sparse numerical vectors based on the importance of words within the corpus. We extract both unigrams and bigrams, enabling the model to capture informative domain-specific phrases such as “blood pressure” and “breast cancer.” The resulting TF-IDF vectors are used to train a multiclass logistic regression classifier implemented using a one-versus-rest scheme. Despite its simplicity, this approach remains a strong baseline due to its efficiency and strong performance on high-dimensional sparse features, providing a transparent and interpretable benchmark against which the benefits of more complex neural models can be evaluated.

2.2 Bag-of-Words

This model applies a Bag-of-Words approach for text preprocessing. The text data was converted into numerical vectors using a text vectorization layer, and model performance was compared across different encoding methods and n-gram configurations. The model architecture consists of a two-layer deep neural network, including a dense layer followed by a dropout layer to improve generalization and reduce overfitting. The output layer uses a softmax activation function to generate probability scores across five patient conditions. To improve model performance, hyperparameter tuning was conducted using Random Search over 20 trials. The tuning process explored different combinations of the number of units in the dense layer, dropout rate, and learning rate to identify the optimal configuration.

2.3 Fine-tuned BERT

We fine-tune a BERT-base model (bert_base_en_uncased) [3] for the medical abstract classification task. The input abstracts are tokenized using the BERT tokenizer and truncated or padded to a maximum sequence length of 256 tokens. The contextual representations produced by the BERT backbone are passed to a classification layer to predict one of five disease categories. The model is trained with the Adam optimizer (learning rate 5e-5) and the Sparse Categorical Cross-Entropy loss function. Training is conducted for 5 epochs with a batch size of 4. To address class imbalance in the dataset, class weights are applied during training.

2.4 Fine-tuned BiomedBERT

Our final model uses BiomedBERT (previously named PubMedBERT) microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext [5], a BERT-base architecture pre-trained exclusively on PubMed abstracts and PubMed Central full-text articles from scratch, without any general-domain text in the pre-training corpus. This is the key distinction from BioBERT, which continues training from a general BERT checkpoint. Because PubMedBERT's vocabulary and pre-training distribution are derived entirely from biomedical text, it is well-suited for the present task, where the test-time data (PubMed abstracts) closely match the pre-training domain [2].

2.5 Evaluation Metrics

All models are evaluated on the same held-out test set of 2,888 samples. We report accuracy, macro-averaged F1, macro precision, and macro recall. Given the class imbalance, macro F1 is used as the primary metric, as it treats each class equally regardless of sample size. We additionally report per-class precision, recall, and F1-score, and visualize performance using confusion matrices.

3. Experimental Setup

3.1 Training Configuration

To ensure consistency, reproducibility, and fair comparison across different models, all experiments were conducted on Google Colab Pro with an A100 GPU. Random seeds were fixed to ensure reproducibility of results. Training hyperparameters for deep learning models are summarized in Table 2.

	BERT	BiomedBERT
Pre-trained model	bert-base-en-uncased	BiomedNLP-BiomedBERT
Optimizer	Adam	AdamW
Batch size	4	32
Epochs	5	3
Learning rate	5e-5	3e-5
Max sequence length	256 tokens	512 tokens

Table 2: Training configuration for deep learning models

3.2 Handling Class Imbalance

The *General Pathological Conditions* class comprises 33.28% of the training data, while *Digestive System Diseases* accounts for only 10.35%. Without correction, models tend to over-predict the majority class. We address this issue using model-specific strategies across our experiments:

- **TF-IDF + Logistic Regression:** Balanced class weights are enabled in the logistic regression objective, scaling each sample's loss contribution inversely proportional to its class frequency.
- **BoW:** Balanced class weights are computed, scaling each sample's loss contribution inversely proportional to its class frequency.
- **BERT:** Class weights are applied during training to penalize misclassification of minority classes more heavily.
- **BiomedBERT:** Weighted CrossEntropyLoss with per-class weights: Neoplasms 0.913, Digestive System Diseases 1.933, Nervous System Diseases 1.500, Cardiovascular Diseases 0.946, General Pathological Conditions 0.601.

4. Results

4.1 Overall Performance

Table 3 summarizes the test-set performance of all four models. BiomedBERT achieves the highest macro F1 and accuracy among all models, validating the hypothesis that domain-specific pretraining is beneficial for medical text classification. While the fine-tuned BERT model achieves performance comparable to the TF-IDF baseline, BiomedBERT yields

clear improvements in both macro F1 and accuracy, highlighting the benefit of domain-specific pretraining.

Model	Macro F1	Accuracy	Macro Precision	Macro Recall
TF-IDF + Logistic Regression	0.6020	0.6008	0.5874	0.6267
BoW	0.5574	0.5578	0.5438	0.6196
BERT (fine-tuned)	0.6000	0.5997	0.5900	0.6500
BiomedBERT (fine-tuned)	**0.6535**	**0.6544**	**0.6525**	**0.7218**

Table 3: Test-set performance across all models; macro F1 is the primary metric

The TF-IDF + logistic regression baseline achieves a macro F1 of 0.6020 and accuracy of 0.6008. Despite its simplicity, this model performs competitively, reflecting the strong lexical distinctiveness of medical terminology. N-gram features effectively capture the highly specialized vocabularies of disease categories such as *Neoplasms* and *Cardiovascular Diseases*. As a result, even a linear classifier operating on TF-IDF representations can achieve fairly strong performance without requiring contextual language modeling.

The optimized bag-of-words model uses a text vectorization layer with unigram count encoding for the top 5,000 tokens. The final best model includes 64 units in the dense layer, 0.4 dropout rate, and a learning rate of 0.01. This configuration achieves a macro F1 score of 0.5574 and an accuracy of 0.5578. Compared to the TF-IDF + logistic regression model, this model performs worse. This may be due to the relatively small training dataset, which limits the deep neural network's ability to learn robust patterns. In addition, the Bag-of-Words representation may cause the neural network to overfit to noisy or less informative features in the text.

The fine-tuned BERT model achieves performance comparable to the TF-IDF baseline, with a macro F1 score of approximately 0.60 and a similar accuracy. This suggests that contextual embeddings provide limited additional benefit over n-gram representations for this dataset. Although BERT's bidirectional attention mechanism can capture semantic dependencies and negation patterns that TF-IDF cannot represent, these advantages appear to contribute only marginal improvements in this task. This is consistent with prior observations that BERT's pre-training on general English text offers limited benefits when applied to dense biomedical terminology.

BiomedBERT achieves the best performance across all models, with a macro F1 of 0.6535 and accuracy of 0.6544. Compared to general BERT, BiomedBERT improves macro F1 by 5.4 percentage points and macro recall by 7.2 percentage points. The recall gain is particularly notable: BiomedBERT is substantially better at identifying true positives across all classes, suggesting that its biomedical vocabulary and pre-training distribution help it recognize domain-specific expressions that general BERT misses.

Overall macro F1 and accuracy remain below 70% across all models. This is primarily attributable to the *General Pathological Conditions* class, which consistently achieves very low recall (0.31–0.40) across all models and substantially depresses both metrics. The remaining four classes achieve considerably higher F1 scores of 0.56-0.76, demonstrating that the classification task is well-defined for those categories.

4.2 Per-Class Analysis

Table 4 presents per-class results for all four models side-by-side, enabling direct comparison of where each model excels and where it struggles. Figure 1 further visualizes these results through confusion matrices.

Class	Model	Precision	Recall	F1-Score
Neoplasms	**TF-IDF + LR**	0.6850	0.7694	0.7010
	BoW	0.6652	0.7409	0.7122
	BERT	0.6990	0.7200	0.7000
	BiomedBERT	**0.7141**	**0.8167**	**0.7620**
Digestive System Diseases	**TF-IDF + LR**	0.5000	0.6355	0.5596
	BoW	0.4292	0.6789	0.5259
	BERT	**0.5200**	0.6700	0.5800
	BiomedBERT	0.5120	**0.8595**	**0.6417**
Nervous System Diseases	**TF-IDF + LR**	0.5442	0.6078	0.5742
	BoW	0.4689	0.7247	0.5694
	BERT	0.4900	0.7300	0.5900
	BiomedBERT	**0.5838**	**0.7688**	**0.6637**
Cardiovascular Diseases	**TF-IDF + LR**	0.6587	0.7213	0.6886
	BoW	0.6312	0.7295	0.6768

	BERT	0.6200	0.8400	0.7200
	BiomedBERT	**0.6767**	**0.8508**	**0.7538**
General Pathological Conditions	**TF-IDF + LR**	0.5494	**0.3996**	**0.4627**
	BoW	0.5244	0.2237	0.3136
	BERT	0.6400	0.3000	0.4100
	BiomedBERT	**0.7758**	0.3132	0.4463

Table 4: Test-set class-level precision, recall, and F1-score for models

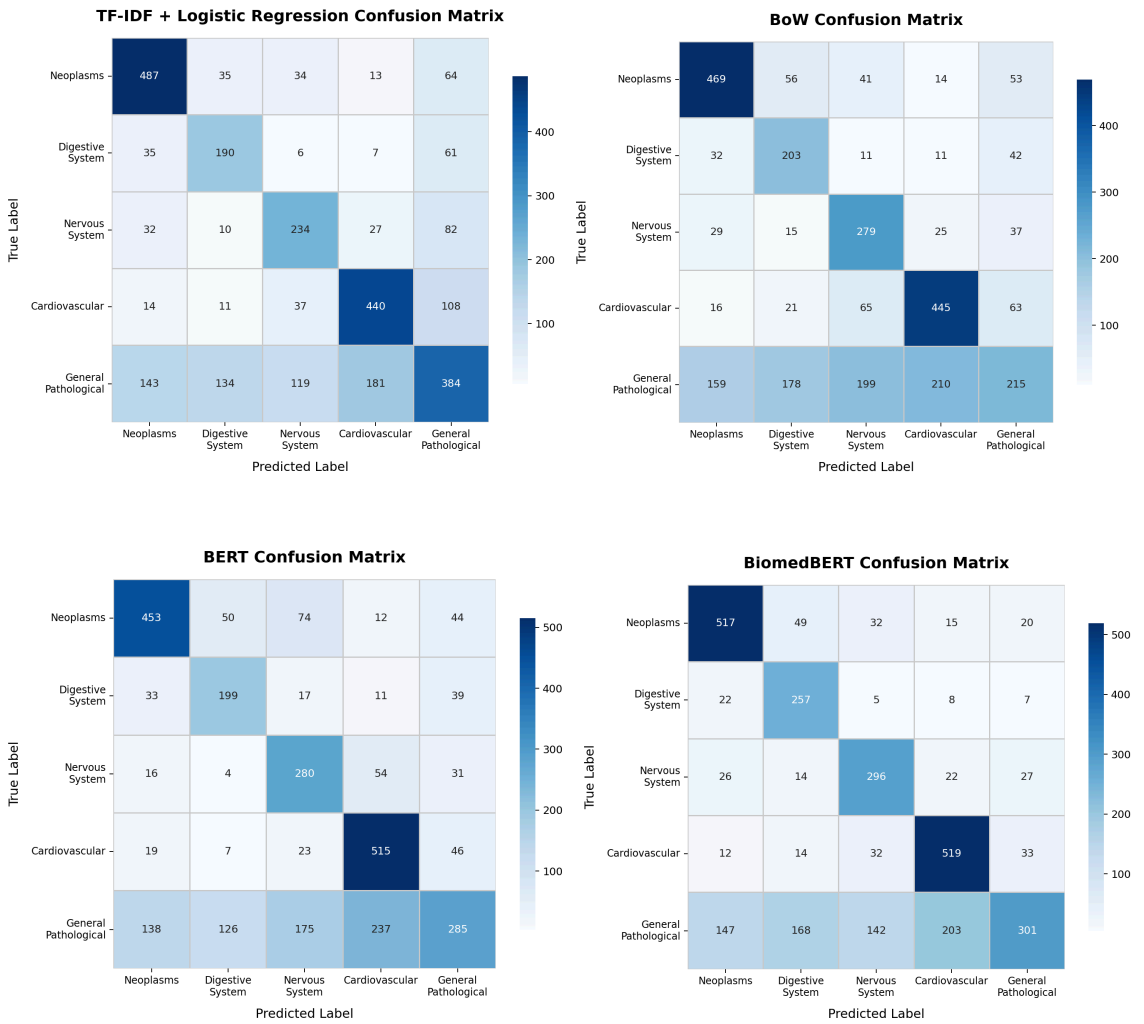

Figure 1: Test-set confusion matrices for models (TF-IDF + logistic regression, bag-of-words, BERT, and BiomedBERT)

Neoplasms and *Cardiovascular Diseases* are the best-classified categories across all models, achieving F1 scores of 0.69-0.76. These classes have highly distinctive clinical vocabularies

(oncology and cardiac terminology) that are well-captured by both n-gram and contextual representations.

BiomedBERT substantially improves recall compared to TF-IDF on minority classes. The most notable improvement is in D*igestive System Diseases*, where recall rises from 0.6355 (TF-IDF) and 0.6789 (BoW) to 0.8595 (BiomedBERT), demonstrating that domain-specific pretraining captures gastrointestinal terminology far more effectively than both classical and general-purpose approaches.

General Pathological Conditions is consistently the weakest class across all four models, with recall ranging from 0.22 to 0.40 and F1 from 0.31 to 0.46. Qualitative inspection of this class reveals a heterogeneous mix of conditions, including prosthetic devices, bronchopleural fistulas, neuropathies, and cutaneous syndromes that overlap semantically with all other categories. This represents a fundamental limitation of the dataset's label schema: the category was defined broadly as a catch-all in the original dataset paper [1], which was designed for unsupervised zero-shot classification benchmarking rather than supervised fine-tuning. The ambiguous boundary of this class limits the discriminative ceiling for any classifier, regardless of architecture.

BiomedBERT shows the highest precision on *General Pathological Conditions* (0.7758 vs. 0.5494 for TF-IDF and 0.4751 for BoW), but at the cost of a low recall (0.3132). This precision-recall tradeoff reflects the effect of the weighted loss combined with the class's inherent ambiguity: the model has learned to predict this class selectively, reducing false positives while missing many true positives.

5. Discussion

BiomedBERT outperformed general BERT on this medical text classification task, consistent with prior work on biomedical NLP [2]. The gain is attributable to two factors: (1) BiomedBERT's vocabulary is derived entirely from biomedical text, reducing out-of-vocabulary fragmentation for clinical terms; (2) its pre-training distribution closely matches the test-time data distribution (PubMed abstracts), reducing the domain shift that general BERT must overcome during fine-tuning.

The TF-IDF and BoW baselines performed surprisingly well relative to the transformer models. This is partly explained by the high lexical discriminativeness of medical terminology: diseases such as *Neoplasms* and *Cardiovascular Conditions* have highly distinctive vocabularies that n-gram models capture effectively. Notably, TF-IDF slightly outperforms BERT on overall Macro F1 (0.6020 vs. 0.6000), suggesting that for datasets with

clearly separable class vocabularies, the marginal benefit of contextual representations may be limited without domain-specific pretraining.

When evaluating model performance, accuracy is a misleading metric for imbalanced classification. A naive classifier that always predicts *General Pathological Conditions* would achieve 33% accuracy, and optimizing for accuracy encourages models to favor this class. However, macro F1 penalizes poor performance on any class equally, making it an appropriate primary metric. This insight motivated our use of macro F1 for both model selection and final reporting.

All models plateau below 70% accuracy, reflecting a fundamental ceiling imposed by the ambiguous *General Pathological Conditions* class rather than any modeling failure. Future work could explore: (1) re-labeling or splitting this category using medical ontologies such as MeSH or ICD-10; (2) applying larger biomedical language models such as BioGPT or Med-PaLM; (3) using ensemble methods that combine the lexical strengths of TF-IDF with the contextual representations of BiomedBERT.

6. Conclusion

This project demonstrates that automated medical abstract classification is a tractable NLP task, and that domain-specific pretraining consistently outperforms general-purpose approaches. Across all four models, BiomedBERT achieved the highest macro F1 of 0.6535, confirming that aligning the pre-training domain with the target task yields measurable gains. The TF-IDF baseline proved surprisingly competitive, highlighting the importance of lexical features in specialized domains such as biomedical literature. Overall performance was constrained by the inherently ambiguous *General Pathological Conditions* class, a dataset-level limitation that no model architecture can fully overcome. These findings underscore the importance of domain-specific pretraining, appropriate evaluation metrics, and careful dataset design when applying NLP to biomedical text classification. More broadly, these results suggest that domain-adapted language models may play an important role in large-scale biomedical information management systems, helping clinicians and researchers navigate the rapidly expanding scientific literature.

References

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[3] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In Proceedings of NAACL-HLT 2019, pp. 4171–4186.

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[5] Microsoft. BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext. Hugging Face Model Hub.https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext.

Appendix (Code)

GitHub Repository: https://github.com/QianyuFan8/HODL_Final_Project
TF-IDF + Logistic Regression:
https://colab.research.google.com/drive/1piAvcwo0imt9j7j0rMF6EYJwIeE3cnDi?usp=sharing
BoW:
https://colab.research.google.com/drive/17Ry8LTf1-4thfDFtm3iz9CMzanst8yZW?usp=sharing
BERT:
https://colab.research.google.com/drive/1NA-0Vx00gKbDyjJBgVevnZ1G5e4VktXG?usp=sharing
BiomedBERT:
https://colab.research.google.com/drive/1MWAOGuPoW0ENr0sQt3Rg0R7HXYQIzOuO?usp=sharing